

WHERE THE REALLY HARD PROBLEMS ARE

Original Paper by Cheeseman et al. (1991)
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LIMITATIONS

The authors tested the Hamilton circuit, graph colouring, K-sat, and the travelling salesman problems. While this is a good sample, further research was needed to establish how generalisable their findings were.

Sleepers et al. [2] critiqued the results of this study [1] vis-a-vis the travelling salesman problem (TSP) as being the result of an issue with how the graphs were generated- thus, this paper did not show that phase transitions exist for the TSP or, by extension, any optimisation problem

Further studies [3] [4] found that, while the highest median difficulty is found in the transition region - the hardest individual problems (and sometimes the highest mean difficulties) are found in the under/over-constrained regions. Something not discussed by Cheeseman et al.

REFERENCES

- [1] "Where The Really Hard Problems Are" Cheeseman et al.
- [2] "Where The Really Hard Problems Aren't" Sleepers et al
- [3] "Easy Problems Are Sometimes Hard" Gent, Walsh
- [4] "The Hardest Constraint Problems: A Double Phase Transition" Hogg, Williams
- [5] "Where The Exceptionally Hard Problems Are" Smith
- [6] "Rigorous location of phase transitions in hard optimization problems" - Achlioptas et al.

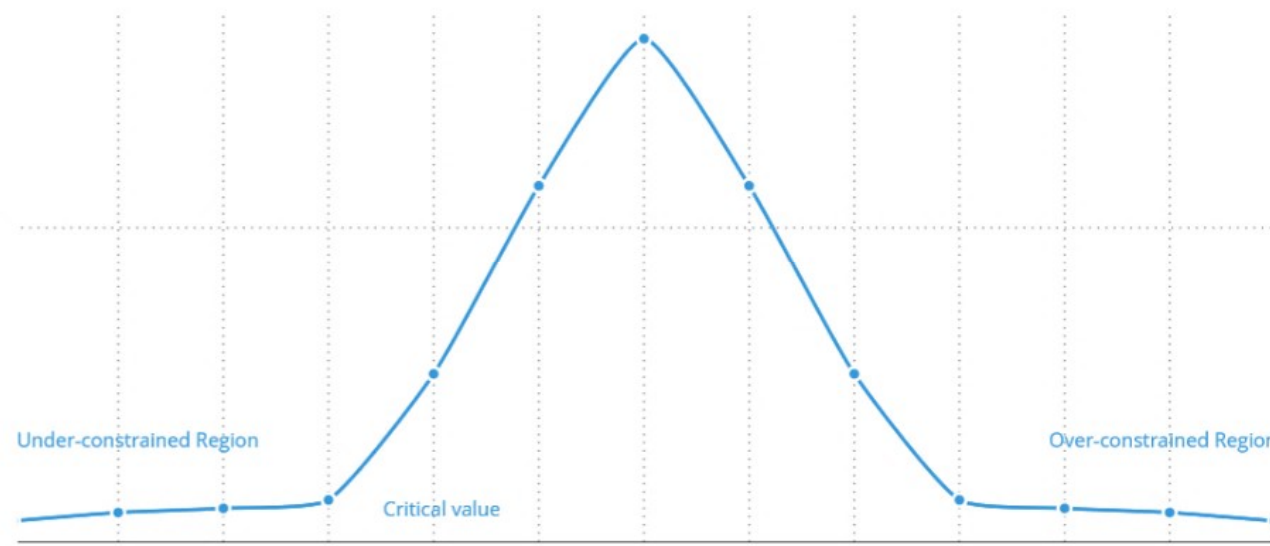
INTRODUCTION

NP-hard problems are considered to be computationally difficult to solve, yet most common instances of these problems are easily solved by sensible algorithms. These problems are "hard" because of the hardest problems in the class, if these problems are uncommon - that begs the question "Where are the hard problems". This paper

found that the hard problems exist in specific regions of the problem space, demarcated by a "critical value" of an "order parameter". The computational difficulty rapidly increases at the critical value, before falling back down to a much easier level. The "hard" problems are found in this region of the problem space. This effect can be seen in the below graph:

Computational Difficulty Vs Order Parameter

For NP Hard Problems



IMPACT ON THE FIELD

This paper had a very significant impact on the fields of constraint programming and computational complexity. It was the first to properly explain why hard problems are often easy to solve in practice - discovering the relationship between difficulty and order parameters. It generated many new research questions which were picked up and investigated by many other authors. Some [3,4,5] were inspired to look

further into the exceptionally difficult problems that do not follow the trend discovered by Cheeseman et al., gaining new insights into algorithmic edge cases. Fourteen years after its publication, papers [6] are still trying to further investigate this phenomenon and define it more rigorously. According to Semantic Scholar, this paper has 1,350 citations at the time of writing, making it an extremely influential work.

RESULTS

Cheeseman et al. found different order parameters for multiple classes of problems which could differentiate between truly "hard" instances and those which are easy to solve.

They discovered that the region of difficult problems is where the odds of a solution existing increase rapidly from near 0% to near 100%. It is in this transition that the truly hard problems exist. This region begins at the "critical value" of the order parameter.

When the problems are underconstrained, there are many solutions, and finding one is easy; when they are overconstrained, it becomes quick to dismiss large areas of the search tree. It is only in the intermediate region where many near-solutions must be checked fully, making the problem computationally difficult.